

- Take a screenshot of the prompt and the dig command produced.

To query the A record of `www.pdx.edu` using the `dig` command, targeting PSU's local DNS server at `131.252.208.53` and using TCP, you can use the following command:

```
bash
```

[Copy code](#)

```
dig @131.252.208.53 www.pdx.edu A +tcp
```

Explanation:

- `dig` : The command-line tool for querying DNS.
- `@131.252.208.53` : Specifies the DNS server to query (PSU's local DNS server in this case).
- `www.pdx.edu` : The domain name for which the A record is being requested.
- `A` : Specifies the type of record being queried (IPv4 address record).
- `+tcp` : Forces the query to use TCP instead of the default UDP.

- Take a screenshot of the records returned for your lab notebook.

ALL ACTIVITY MAY BE RECORDED

=====

```
* CAT Support:    https://cat.pdx.edu/
* Email:         support@cat.pdx.edu
* Phone:         503-725-5420
* Chat:          https://support.cat.pdx.edu
* Location:      FAB 82-01
```

```
jgafron@ada:~$ dig @131.252.208.53 www.pdx.edu A +tcp
```

```
; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> @131.252.208.53 www.pdx.edu A +tcp
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60425
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: dbd4123612a3e5230100000067875d9acf721c51b2103a4d (good)
;; QUESTION SECTION:
;www.pdx.edu.                IN      A

;; ANSWER SECTION:
www.pdx.edu.                 60      IN      A      18.161.6.120
www.pdx.edu.                 60      IN      A      18.161.6.84
www.pdx.edu.                 60      IN      A      18.161.6.96
www.pdx.edu.                 60      IN      A      18.161.6.112

;; Query time: 67 msec
;; SERVER: 131.252.208.53#53(131.252.208.53) (TCP)
;; WHEN: Tue Jan 14 23:02:51 PST 2025
;; MSG SIZE rcvd: 132
```

```
jgafron@ada:~$ dig @131.252.208.53 www.pdx.edu MX +tcp
```

```
; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> @131.252.208.53 www.pdx.edu MX +tcp
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 40401
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 6b180b52d890cfb00100000067875e441080f553e78da901 (good)
;; QUESTION SECTION:
;www.pdx.edu.                IN      MX

;; AUTHORITY SECTION:
www.pdx.edu.                 900     IN      SOA     ns-988.awsdns-59.net. awsdns-hostmaster.amazon.com. 1 7200 900 1
209600 86400

;; Query time: 42 msec
;; SERVER: 131.252.208.53#53(131.252.208.53) (TCP)
;; WHEN: Tue Jan 14 23:05:41 PST 2025
;; MSG SIZE rcvd: 152
```

- What cloud provider hosts the web site for www.pdx.edu?
- What cloud provider handles mail for pdx.edu?

Google LLC

- Take a screenshot of the results for both records for your lab notebook.

```
jgafron@ada:~$ dig @walt.ee.pdx.edu mashimaro.cs.pdx.edu A

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> @walt.ee.pdx.edu mashimaro.cs.pdx.edu A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 1206
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 60174a44c633aa3e010000006788278c4aa86b7e5472c653 (good)
;; QUESTION SECTION:
;mashimaro.cs.pdx.edu.          IN      A

;; ANSWER SECTION:
mashimaro.cs.pdx.edu.  14400   IN      A      131.252.220.66

;; Query time: 0 msec
;; SERVER: 131.252.208.38#53(walt.ee.pdx.edu) (UDP)
;; WHEN: Wed Jan 15 13:24:29 PST 2025
;; MSG SIZE rcvd: 93
```

```
jgafron@ada:~$ dig NS mashimaro.cs.pdx.edu

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> NS mashimaro.cs.pdx.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 45043
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;mashimaro.cs.pdx.edu.          IN      NS

;; AUTHORITY SECTION:
cs.pdx.edu. 300 IN SOA walt.ee.pdx.edu. support.cat.pdx.edu. 2025011502 600 300 1209600 300

;; Query time: 5 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Jan 15 13:22:08 PST 2025
;; MSG SIZE rcvd: 105
```

- List all of the iterative dig commands performed for the lookup

```
jgafron@ada:~$ dig f.root-servers.net A

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> f.root-servers.net A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51046
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:: udp: 65494
;; QUESTION SECTION:
f.root-servers.net.          IN      A

;; ANSWER SECTION:
f.root-servers.net.          99      IN      A          192.5.5.241

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Jan 15 14:10:33 PST 2025
;; MSG SIZE rcvd: 63
```

S

```
jgafron@ada:~$ dig @127.0.0.53 com NS

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> @127.0.0.53 com NS
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25326
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 27

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;com.                IN      NS

;; ANSWER SECTION:
com.                 5167    IN      NS      d.gtld-servers.net.
com.                 5167    IN      NS      m.gtld-servers.net.
com.                 5167    IN      NS      a.gtld-servers.net.
com.                 5167    IN      NS      i.gtld-servers.net.
com.                 5167    IN      NS      h.gtld-servers.net.
com.                 5167    IN      NS      b.gtld-servers.net.
com.                 5167    IN      NS      c.gtld-servers.net.
com.                 5167    IN      NS      e.gtld-servers.net.
com.                 5167    IN      NS      g.gtld-servers.net.
com.                 5167    IN      NS      l.gtld-servers.net.
com.                 5167    IN      NS      j.gtld-servers.net.
com.                 5167    IN      NS      f.gtld-servers.net.
com.                 5167    IN      NS      k.gtld-servers.net.
```

```
jgafron@ada:~$ dig d.gtld-servers.net A NS
;; Warning, extra type option

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> d.gtld-servers.net A NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17424
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;d.gtld-servers.net.      IN      NS

;; AUTHORITY SECTION:
gtld-servers.net.        10800   IN      SOA      av4.nstld.com. nstld.verisign-
grs.com. 1736900310 3600 900 1209600 86400

;; Query time: 21 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Jan 15 14:13:30 PST 2025
;; MSG SIZE rcvd: 115
```

```
jgafron@ada:~$ dig @d.gtld-servers.net google.com NS

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> @d.gtld-servers.net google.com
NS
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 27164
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 9
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;google.com.                IN      NS

;; AUTHORITY SECTION:
google.com.                172800  IN      NS      ns2.google.com.
google.com.                172800  IN      NS      ns1.google.com.
google.com.                172800  IN      NS      ns3.google.com.
google.com.                172800  IN      NS      ns4.google.com.

;; ADDITIONAL SECTION:
ns2.google.com.            172800  IN      AAAA     2001:4860:4802:34::a
ns2.google.com.            172800  IN      A        216.239.34.10
```

```
jgafron@ada:~$ dig +tcp @ns1.google.com console.cloud.google.com A

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> +tcp @ns1.google.com console.c
loud.google.com A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 46050
;; flags: qr aa rd; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;console.cloud.google.com.  IN      A

;; ANSWER SECTION:
console.cloud.google.com. 300     IN      CNAME    www3.l.google.com.
www3.l.google.com.       300     IN      A        142.250.69.206

;; Query time: 27 msec
;; SERVER: 216.239.32.10#53(ns1.google.com) (TCP)
;; WHEN: Wed Jan 15 14:17:32 PST 2025
```

Commands used:

1. Dig

2. Dig f.root-servers.net A
3. Dig @127.0.0.53 com NS
4. Dig @d.gtld-servers.net google.com NS
5. Dig +tcp @ns1.google.com cloud.google.com A

- Take a screenshot of the prompt and the command produced



Produce a single command using dig egrep and awk that list all of the ipv4 addresses that espn.go.com points to

Here's the single command that lists all IPv4 addresses that `espn.go.com` points to using `dig`, `egrep`, and `awk`:

bash

Copy Edit

```
dig espn.go.com +short | egrep '^([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+)$' | awk '{print $1}'
```

- Take a screenshot of its results for your lab notebook

```
jgafron@ada:~$ d
99.84.66.55
99.84.66.98
99.84.66.108
99.84.66.17
```

- Take a screenshot of the command and its results for your lab notebook

```
jgafron@ada:~$ for ip in $(dig espn.go.com +short | egrep '^([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+)$'); do
  dig -x $ip +short
done | egrep -v '^$' | awk '{print $1}'
server-99-84-66-108.hio50.r.cloudfront.net.
server-99-84-66-98.hio50.r.cloudfront.net.
server-99-84-66-17.hio50.r.cloudfront.net.
server-99-84-66-55.hio50.r.cloudfront.net.
```

- Take a screenshot of the results in your lab notebook

```

jgafron@ada:~$ cat 220hosts.txt
colt45.cs.pdx.edu.
kingcobra.cs.pdx.edu.
mickeys.cs.pdx.edu.
magnum.cs.pdx.edu.
phatboy.cs.pdx.edu.
schlitz.cs.pdx.edu.
boar.cs.pdx.edu.
dog.cs.pdx.edu.
dragon.cs.pdx.edu.
horse.cs.pdx.edu.
monkey.cs.pdx.edu.
ox.cs.pdx.edu.
rabbit.cs.pdx.edu.
rat.cs.pdx.edu.
rooster.cs.pdx.edu.
sheep.cs.pdx.edu.
snake.cs.pdx.edu.
tiger.cs.pdx.edu.
assault.cs.pdx.edu.
aztec.cs.pdx.edu.
backalley.cs.pdx.edu.
cbbble.cs.pdx.edu.
dust.cs.pdx.edu.
estate.cs.pdx.edu.
havana.cs.pdx.edu.
inferno.cs.pdx.edu.
italy.cs.pdx.edu.
militia.cs.pdx.edu.
nuke.cs.pdx.edu.
office.cs.pdx.edu.
oilrig.cs.pdx.edu.
piranesi.cs.pdx.edu.
prodigy.cs.pdx.edu.

```

- What geographic locations do ipinfo.io and DB-IP return?

Portland and Blacksburg, Virginia

Geolocation data from		IP2Location	Product: DB6, 2025-1-1
	IP ADDRESS: 131.252.208.53		ISP: Portland State University
	COUNTRY: United States 		ORGANIZATION: Not available
	REGION: Oregon		LATITUDE: 45.5213
	CITY: Portland		LONGITUDE: -122.6859

Geolocation data from		IP2Location	Product: DB6, 2025-1-1
	IP ADDRESS:	198.82.247.66	 ISP: Virginia Polytechnic Institute and State Univ.
	COUNTRY:	United States 	 ORGANIZATION: Not available
	REGION:	Virginia	 LATITUDE: 37.2557
	CITY:	Blacksburg	 LONGITUDE: -80.4315

- Record one address for www.google.com from each result for your lab notebook.

142.251.167.147 for address 192.82.247.66 (Blacksburg Virginia)

142.251.215.228 for address 131.252.208.53 (Portland)

- What are the geographic coordinates of each DNS server and the IP address it resolves for www.google.com?

192.82.247.66 (Blacksburg Virginia) ----->142.251.167.147 (Ashburn Virginia)

131.252.208.53 (Portland Oregon) --> 142.251.215.228 (Seattle Washington)

- Take a screenshot of the results for your lab notebook.

```
jgafron@ada:~$ traceroute 131.252.208.53
traceroute to 131.252.208.53 (131.252.208.53), 30 hops max, 60 byte packets
 1  rdns.cat.pdx.edu (131.252.208.53)  0.781 ms  0.688 ms  0.630 ms
 2  198.82.247.66 (198.82.247.66)  5.919 ms  11.304 ms  5.768 ms
 3  glados.cat.pdx.edu (131.252.208.21)  0.129 ms  0.111 ms  0.152 ms
 4  router.seas.pdx.edu (10.208.91.1)  5.208 ms  5.145 ms  5.078 ms
 5  CORE1.net.pdx.edu (131.252.5.142)  0.523 ms  0.713 ms  0.466 ms
 6  131.252.5.213 (131.252.5.213)  1.579 ms  1.593 ms  1.809 ms
 7  e0-28.switch4.pdx1.he.net (216.218.230.89)  1.718 ms  1.875 ms  100ge0-28.core1.pdx3.he.net (184.104.188.77)  1.447 ms
 8  100ge0-28.core1.pdx3.he.net (184.104.188.77)  1.584 ms  ael1.bas4.por1.sp.lumen.tech (4.68.38.101)  18.012 ms  100ge0-28.core1.pdx3.he.net (184.104.188.77)  1.924 ms
 9  ael1.bas4.por1.sp.lumen.tech (4.68.38.101)  17.924 ms  16.976 ms  16.850 ms
10  RADWARE-LTD.edge9.SanJose1.Level3.net (4.53.29.46)  16.491 ms  16.373 ms  ael1.3502.edgel.SanJose1.net.lumen.tech (4.69.143.14)  20.560 ms
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * 128.173.0.214 (128.173.0.214)  79.991 ms  79.993 ms
23  cas-core.100.2000.cns.vt.edu (198.82.1.143)  79.937 ms  128.173.0.214 (128.173.0.214)  80.054 ms  79.987 ms
24  cas-core.100.2000.cns.vt.edu (198.82.1.143)  79.915 ms  79.846 ms  jeru.cns.vt.edu (198.82.247.66)  79.669 ms
```

```
jgafron@ada:~$ traceroute 142.251.167.147
traceroute to 142.251.167.147 (142.251.167.147), 30 hops max, 60 byte packets
 1 glados.cat.pdx.edu (131.252.208.21)  3.508 ms  3.443 ms  0.781 ms
 2 0015-opsense.cat.pdx.edu (10.208.91.1)  0.136 ms  0.097 ms  0.085 ms
 3 CORE1.net.pdx.edu (131.252.5.142)  5.105 ms  5.114 ms  5.087 ms
 4 131.252.5.213 (131.252.5.213)  0.423 ms  0.409 ms  0.637 ms
 5 google.nmax.net (198.32.195.34)  4.415 ms  4.769 ms  4.298 ms
 6 108.170.255.123 (108.170.255.123)  5.193 ms  192.178.105.35 (192.178.105.35)  4.243 ms  4.730 ms
 7 192.178.105.144 (192.178.105.144)  5.376 ms  192.178.105.64 (192.178.105.64)  3.974 ms  192.178.105.144 (192.178.105.144)  5.501 ms
 8 142.251.64.0 (142.251.64.0)  10.131 ms  10.138 ms  9.899 ms
 9 142.251.226.159 (142.251.226.159)  52.719 ms  52.810 ms  142.251.226.161 (142.251.226.161)  52.727 ms
10 192.178.81.228 (192.178.81.228)  65.469 ms  192.178.81.226 (192.178.81.226)  67.093 ms  192.178.81.228 (192.178.81.228)  65.313 ms
11 142.250.209.246 (142.250.209.246)  65.148 ms  142.250.211.50 (142.250.211.50)  66.960 ms  142.251.244.142 (142.251.244.142)  66.811 ms
12 142.250.211.27 (142.250.211.27)  65.749 ms  142.250.235.95 (142.250.235.95)  67.258 ms  142.250.211.25 (142.250.211.25)  65.575 ms
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 ww-in-f147.1e100.net (142.251.167.147)  65.248 ms  65.808 ms  65.806 ms

jgafron@ada:~$ traceroute 142.251.215.228
traceroute to 142.251.215.228 (142.251.215.228), 30 hops max, 60 byte packets
 1 glados.cat.pdx.edu (131.252.208.21)  3.996 ms  2.443 ms  3.802 ms
 2 0015-opsense.cat.pdx.edu (10.208.91.1)  0.192 ms  0.104 ms  0.114 ms
 3 CORE1.net.pdx.edu (131.252.5.142)  1.013 ms  0.960 ms  0.881 ms
 4 131.252.5.213 (131.252.5.213)  0.551 ms  0.484 ms  0.416 ms
 5 google.nmax.net (198.32.195.34)  4.347 ms  6.014 ms  5.887 ms
 6 108.170.255.123 (108.170.255.123)  6.029 ms  192.178.105.35 (192.178.105.35)  4.507 ms  108.170.255.123 (108.170.255.123)  5.746 ms
 7 142.251.241.137 (142.251.241.137)  4.744 ms  4.753 ms  4.520 ms
 8 sea09s35-in-f4.1e100.net (142.251.215.228)  4.454 ms  4.731 ms  4.212 ms
```

Take a screenshot of the bytes in the packet dump window as shown below

No.		Source	Destination	Protocol	Length	Info
1231	7.426391663	10.138.0.2	99.79.7.93	ICMP	82	Echo (ping) request id=
1639	9.814486400	10.138.0.2	74.125.142.105	ICMP	98	Echo (ping) request id=
1640	9.817020507	74.125.142.105	10.138.0.2	ICMP	98	Echo (ping) reply id=
1794	10.815998155	10.138.0.2	74.125.142.105	ICMP	98	Echo (ping) request id=
1795	10.816367692	74.125.142.105	10.138.0.2	ICMP	98	Echo (ping) reply id=
1999	11.818787298	10.138.0.2	74.125.142.105	ICMP	98	Echo (ping) request id=
2000	11.819182708	74.125.142.105	10.138.0.2	ICMP	98	Echo (ping) reply id=

```

▶ Frame 1230: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface ens4, id 0
▼ Ethernet II, Src: 42:01:0a:8a:00:01 (42:01:0a:8a:00:01), Dst: 42:01:0a:8a:00:02 (42:01:0a:8a:00:02)
  ▶ Destination: 42:01:0a:8a:00:02 (42:01:0a:8a:00:02)
  ▶ Source: 42:01:0a:8a:00:01 (42:01:0a:8a:00:01)
  ▶ Type: 802.3 (600000)

```

- Does the destination MAC address correspond to an interface on the VM, an interface on the default router or an interface on Google's web site?

Yes! The destination MAC address corresponds to the MAC address of the default gateway.

- Does the destination MAC address correspond to an interface on the VM, an interface on the default router or an interface on Google's web site?

The destination MAC of the reply is the MAC address of my wireless interface, or rather the wireless interface of my VM.

- Take a screenshot of the all of the packets returned within Wireshark that includes their packet numbers

No.	Time	Source	Destination	Protocol	Length	Info
223	5.781693	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
224	5.781712	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
280	11.781342	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
281	11.781360	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
349	17.780994	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
350	17.781013	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
472	23.781147	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
473	23.781168	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
701	29.781344	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
702	29.781365	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
963	35.781406	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
964	35.781422	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
1096	41.780875	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
1097	41.780895	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
1215	47.781612	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
1216	47.781629	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
1341	53.781549	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
1342	53.781570	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02
1349	55.107875	10.138.0.2	169.254.169.254	DNS	112	Standard query 0x2336 A course-vm.c.cloud-gafron-jgafron.internal OPT
1350	55.108001	10.138.0.2	169.254.169.254	DNS	112	Standard query 0x140f AAAA course-vm.c.cloud-gafron-jgafron.internal OPT
1351	55.112649	169.254.169.254	10.138.0.2	DNS	201	Standard query response 0x140f AAAA course-vm.c.cloud-gafron-jgafron.internal SOA ns.global.gcedns-prod.internal OPT
1352	55.112715	169.254.169.254	10.138.0.2	DNS	128	Standard query response 0x2336 A course-vm.c.cloud-gafron-jgafron.internal A 10.138.0.2 OPT
1353	55.131295	42:01:0a:8a:00:02	Broadcast	ARP	42	Who has 10.138.0.1? Tell 10.138.0.2
1354	55.133143	42:01:0a:8a:00:01	42:01:0a:8a:00:02	ARP	42	10.138.0.1 is at 42:01:0a:8a:00:01
1356	55.136921	10.138.0.2	169.254.169.254	DNS	92	Standard query 0xaf46 A jgafron.oregonctf.org OPT
1357	55.137031	10.138.0.2	169.254.169.254	DNS	92	Standard query 0x377e AAAA jgafron.oregonctf.org OPT
1363	55.207041	169.254.169.254	10.138.0.2	DNS	174	Standard query response 0x377e AAAA jgafron.oregonctf.org SOA ns-cloud-dl.googledomains.com OPT
1370	55.257386	169.254.169.254	10.138.0.2	DNS	108	Standard query response 0xaf46 A jgafron.oregonctf.org A 35.233.233.233 OPT
1371	55.257963	10.138.0.2	35.233.233.233	TCP	74	51078 → 80 [SYN] Seq=0 Win=65320 Len=0 MSS=1420 SACK_PERM=1 TSval=3790481243 TSecr=0 WS=128
1372	55.262552	35.233.233.233	10.138.0.2	TCP	74	80 → 51078 [SYN, ACK] Seq=0 Ack=1 Win=64768 Len=0 MSS=1420 SACK_PERM=1 TSval=1369565864 TSecr=3790481243 WS=128
1373	55.262646	10.138.0.2	35.233.233.233	TCP	66	51078 → 80 [ACK] Seq=1 Ack=1 Win=65408 Len=0 TSval=3790481248 TSecr=1369565864
1374	55.262774	10.138.0.2	35.233.233.233	HTTP	202	GET / HTTP/1.1
1375	55.263393	35.233.233.233	10.138.0.2	TCP	66	80 → 51078 [ACK] Seq=1 Ack=137 Win=64640 Len=0 TSval=1369565867 TSecr=3790481248
1376	55.263634	35.233.233.233	10.138.0.2	TCP	4290	80 → 51078 [ACK] Seq=137 Ack=137 Win=64640 Len=4224 TSval=1369565867 TSecr=3790481248 [TCP segment of a reassembled PDU]
1377	55.263654	10.138.0.2	35.233.233.233	TCP	66	51078 → 80 [ACK] Seq=137 Ack=4225 Win=61184 Len=0 TSval=3790481249 TSecr=1369565867
1378	55.263660	35.233.233.233	10.138.0.2	TCP	2882	80 → 51078 [PSH, ACK] Seq=4225 Ack=137 Win=64640 Len=2816 TSval=1369565867 TSecr=3790481248 [TCP segment of a reassembled PDU]
1379	55.263660	35.233.233.233	10.138.0.2	HTTP	799	HTTP/1.1 200 OK (text/html)
1380	55.263671	10.138.0.2	35.233.233.233	TCP	66	51078 → 80 [ACK] Seq=137 Ack=7041 Win=58368 Len=0 TSval=3790481249 TSecr=1369565867
1381	55.263676	10.138.0.2	35.233.233.233	TCP	66	51078 → 80 [ACK] Seq=137 Ack=7774 Win=57728 Len=0 TSval=3790481249 TSecr=1369565867
1382	55.264829	10.138.0.2	35.233.233.233	TCP	66	51078 → 80 [FIN, ACK] Seq=137 Ack=7774 Win=58368 Len=0 TSval=3790481250 TSecr=1369565867
1383	55.265394	35.233.233.233	10.138.0.2	TCP	66	80 → 51078 [FIN, ACK] Seq=7774 Ack=138 Win=64640 Len=0 TSval=1369565869 TSecr=3790481250
1384	55.265416	10.138.0.2	35.233.233.233	TCP	66	51078 → 80 [ACK] Seq=138 Ack=7775 Win=58368 Len=0 TSval=3790481251 TSecr=1369565869
1504	57.767082	169.254.169.254	10.138.0.2	HTTP/2...	1832	[TCP ACKed unseen segment] HTTP/1.1 200 OK , JavaScript Object Notation (application/json)
1506	57.767928	10.138.0.2	169.254.169.254	HTTP	281	GET /computeMetadata/v1/?alt=json&last_etag=90b59939a16923d1&recursive=true&timeout_sec=60&wait_for_change=true HTTP/1.1
1507	57.767928	169.254.169.254	10.138.0.2	HTTP/2...	1332	[TCP ACKed unseen segment] HTTP/1.1 200 OK , JavaScript Object Notation (application/json)
1534	58.767023	10.138.0.2	169.254.169.254	HTTP	281	GET /computeMetadata/v1/?recursive=true&alt=json&wait_for_change=true&last_etag=90b59939a16923d1&timeout_sec=60 HTTP/1.1
1552	59.781601	42:01:0a:8a:00:01	Broadcast	ARP	42	Who has 10.138.0.2? Tell 10.138.0.1
1553	59.781620	42:01:0a:8a:00:02	42:01:0a:8a:00:01	ARP	42	10.138.0.2 is at 42:01:0a:8a:00:02

- What packet numbers in the trace are the result of the VM attempting to get the hardware address of the default router?

1353 and 1354

- What is this hardware address?

42:01:0a:8a:00:02

- What packet numbers in the trace correspond to the DNS request for the web site?

1356,1357,1363,1370. These are the DNS protocol packets of my resolver trying to get the records for jgafron.oregon.ctf

- What is the IP address of the local DNS server being queried?

169.254.169.254

TCP

- What packet numbers in the trace correspond to the initial TCP handshake for the web request?

Packets 1371, 1372, and 1373. Although there are a lot of TCP packets here, the actual handshake happens in the first three packets.

- How long does it take to perform the initial TCP handshake?

Less than a second. The first packet of the handshake (the SYN) was sent at "55.257963" and the ACK packet that signalled the end of the handshake, was sent at "55.262646".

HTTP

- What packet numbers in the trace correspond to the actual HTTP request and response?

Packet 1374 is the HTTP request, and Packet 1379 is the response

- How long does it take to process the HTTP request after the handshake?

Less than a second as well. The HTTP request was sent at "55.262774", and the success code/response was sent at "55.263660"